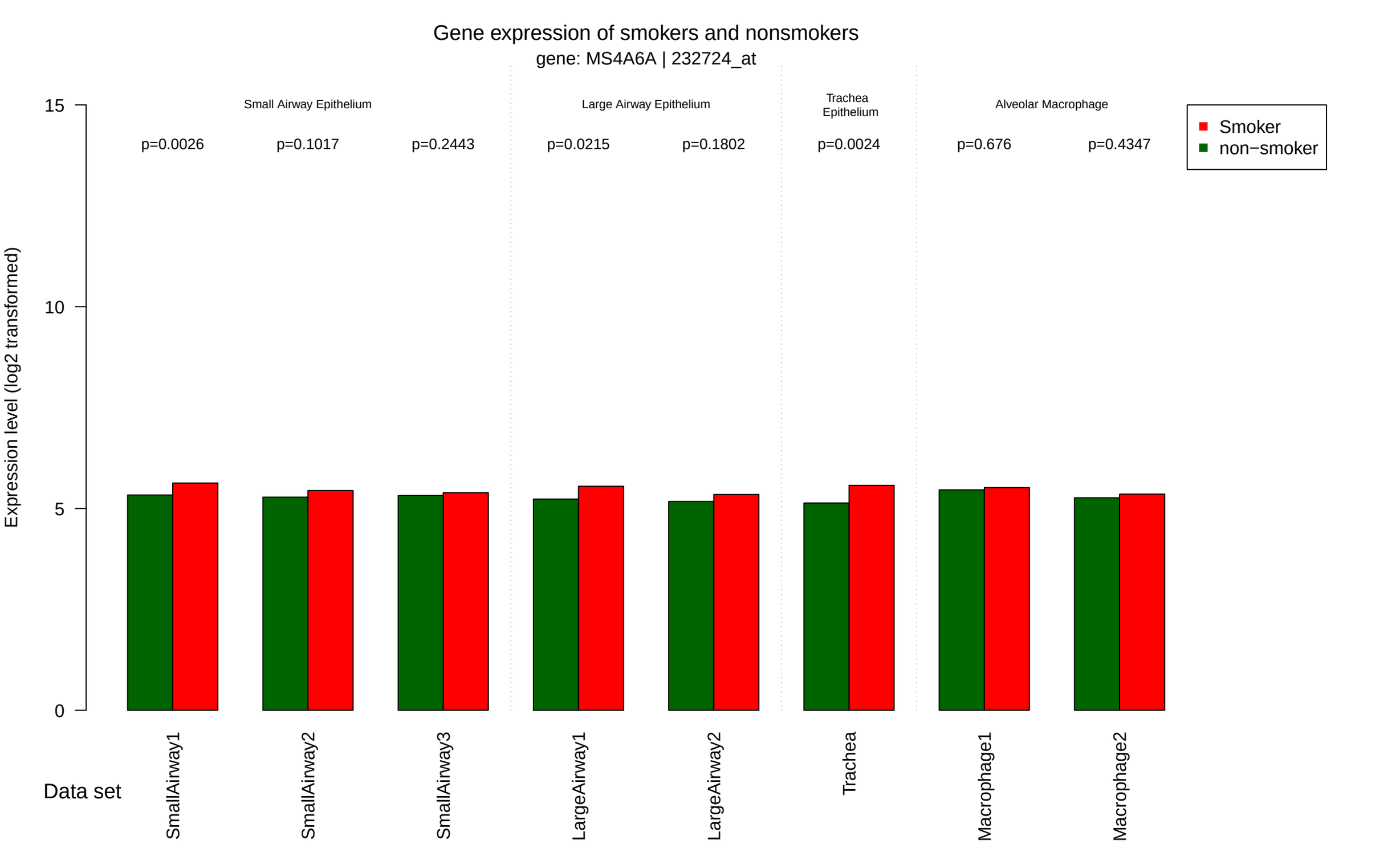
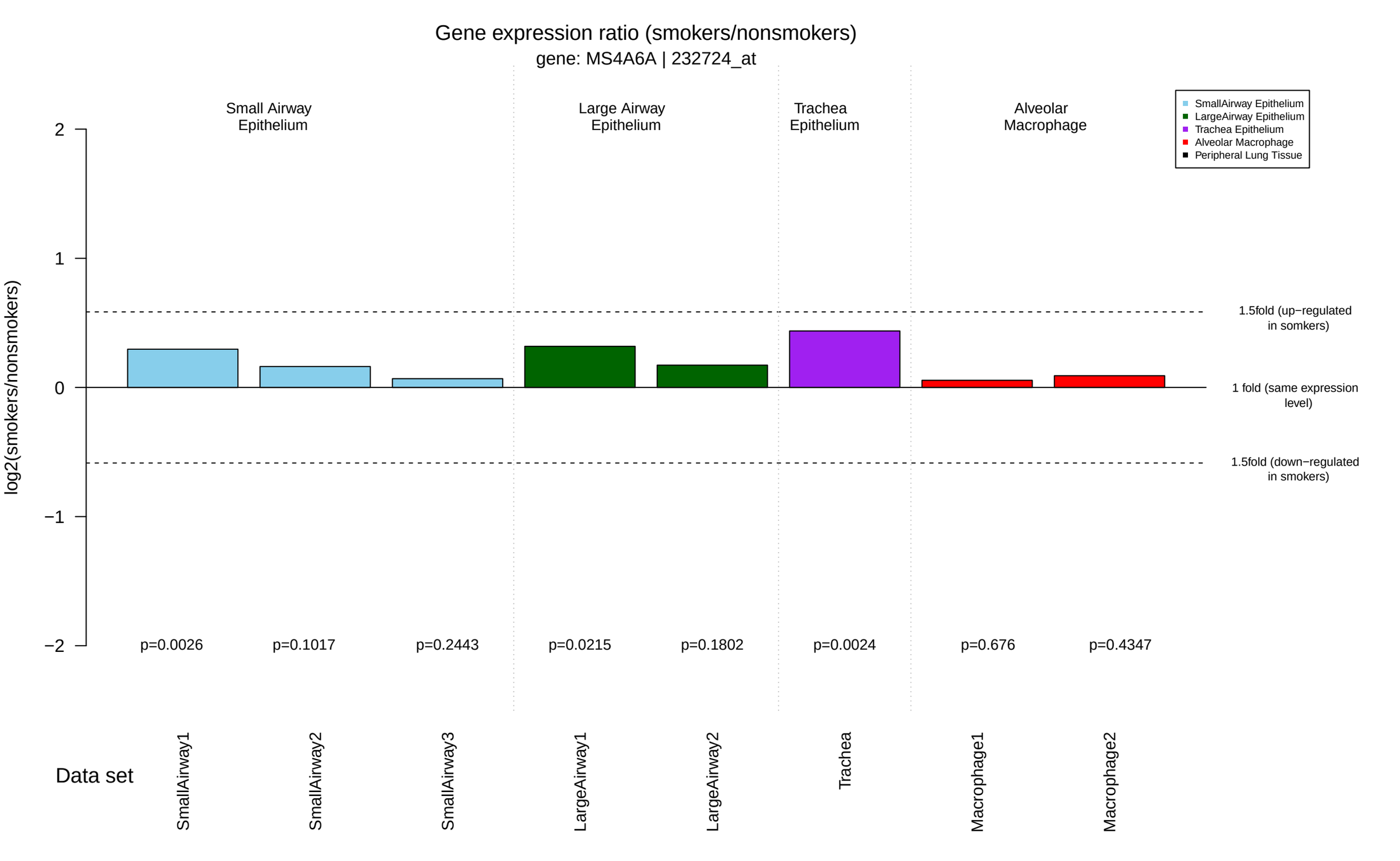
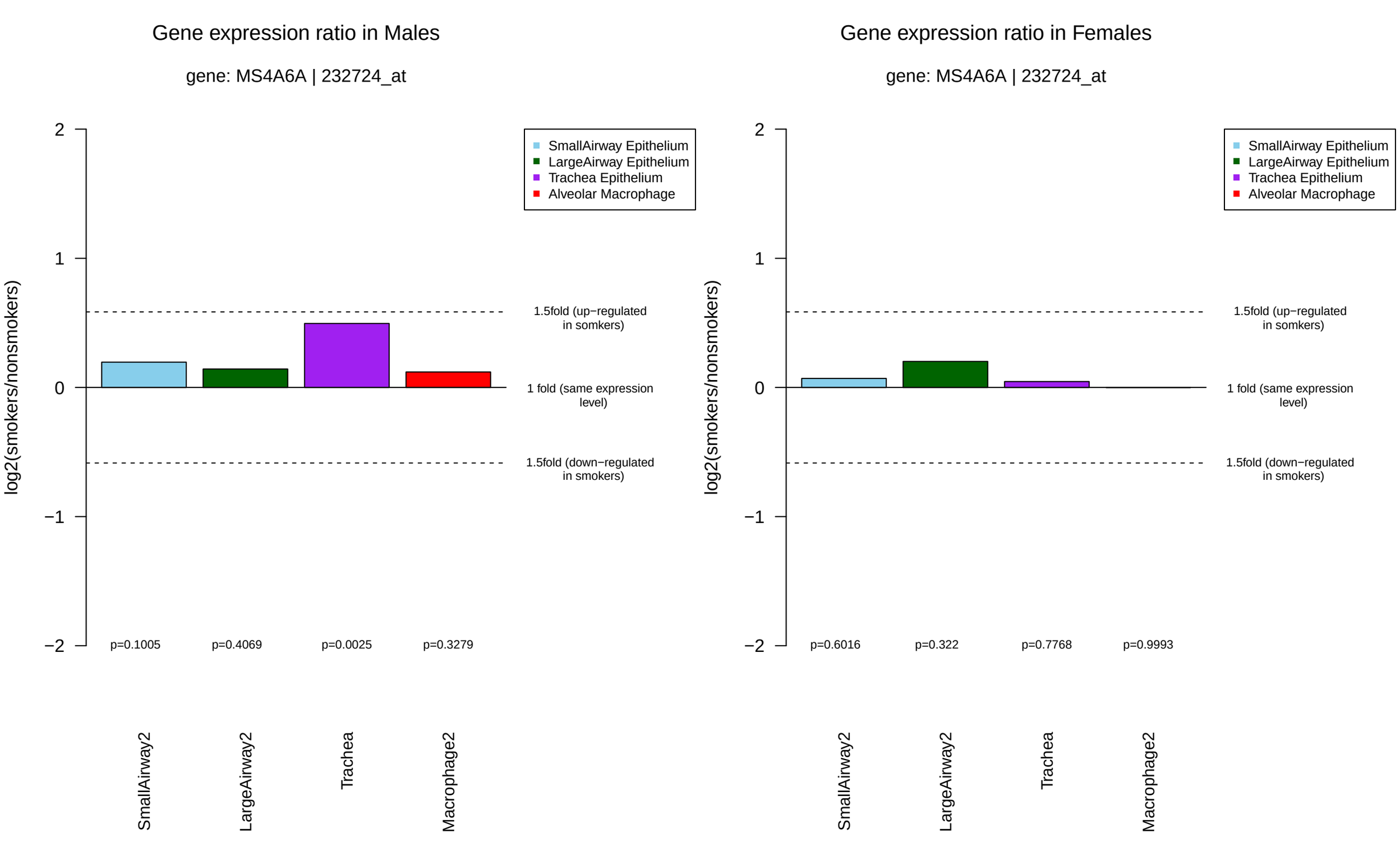
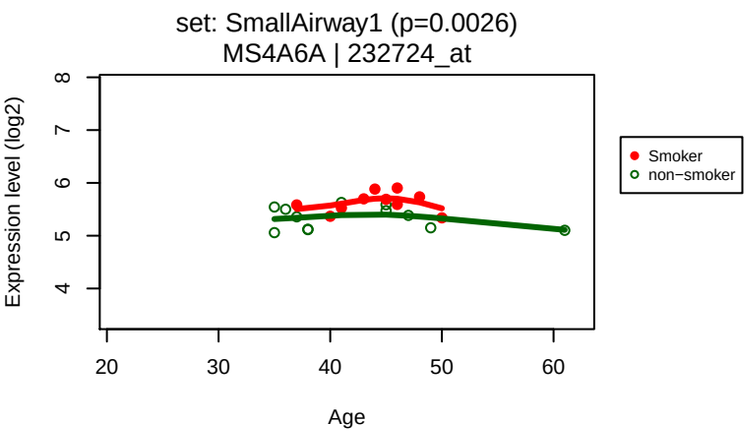
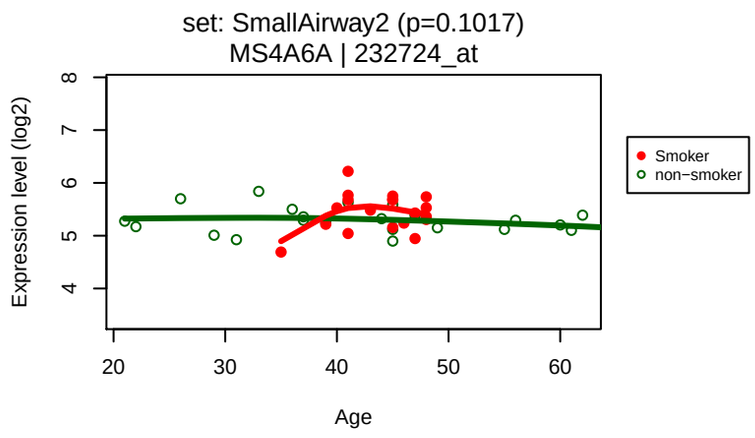
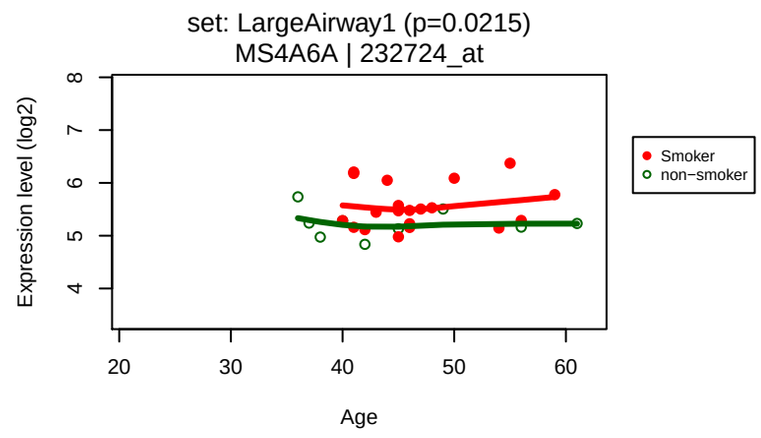

Gene expression of smokers and nonsmokers
gene: MS4A6A | 232724_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0026
p=0.1017
p=0.2443
p=0.0215
p=0.1802
p=0.0024
p=0.676
p=0.4347
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

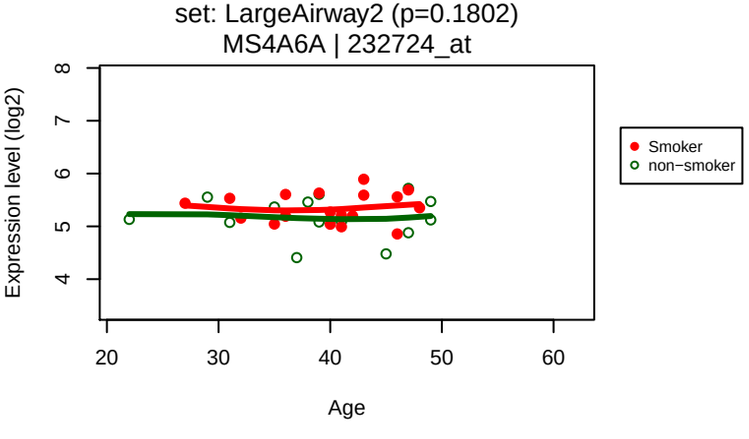

Gene expression ratio (smokers/nonsmokers)
gene: MS4A6A | 232724_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0026
p=0.1017
p=0.2443
p=0.0215
p=0.1802
p=0.0024
p=0.676
p=0.4347
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

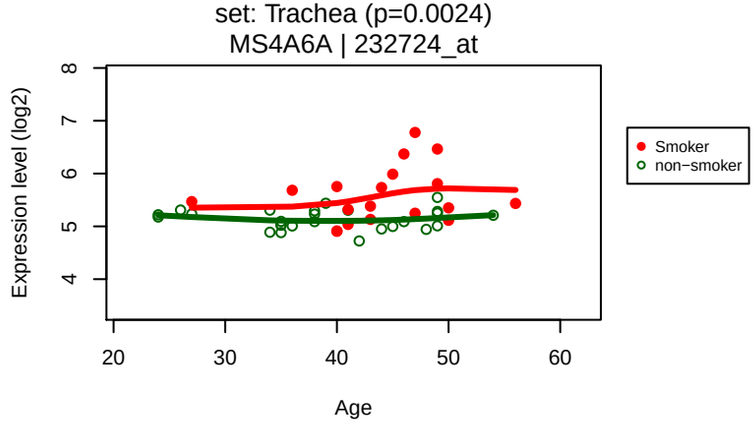

Gene expression ratio in Males
gene: MS4A6A | 232724_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1005
p=0.4069
p=0.0025
p=0.3279
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MS4A6A | 232724_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6016
p=0.322
p=0.7768
p=0.9993
SmallAirway2
LargeAirway2
Trachea
Macrophage2

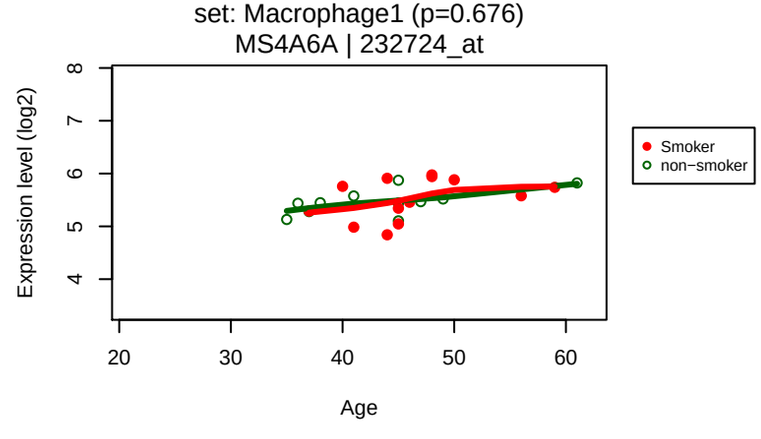

set: SmallAirway1 (p=0.0026)
MS4A6A | 232724_at
Expression level (log2)
Age
Smoker
non−smoker

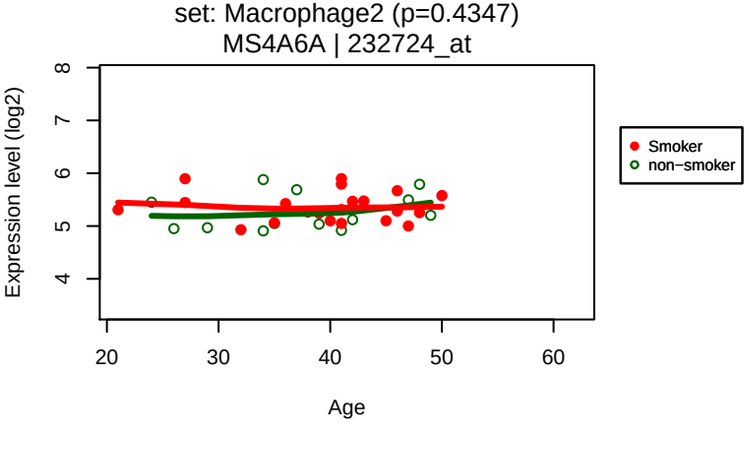

set: SmallAirway2 (p=0.1017)
MS4A6A | 232724_at
Expression level (log2)
Age
Smoker
non−smoker

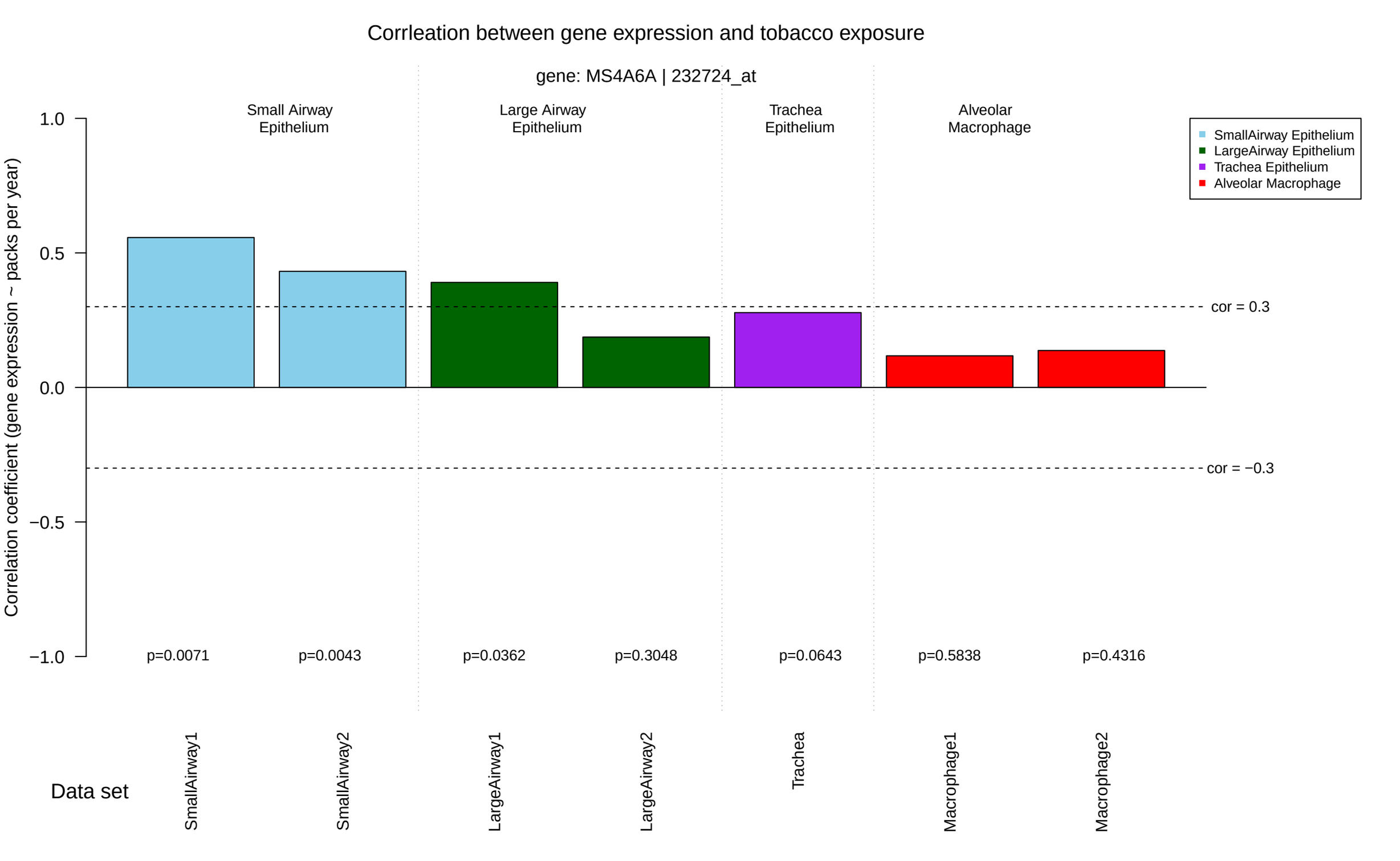

set: LargeAirway1 (p=0.0215)
MS4A6A | 232724_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1802)
MS4A6A | 232724_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0024)
MS4A6A | 232724_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.676)
MS4A6A | 232724_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4347)
MS4A6A | 232724_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MS4A6A | 232724_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0071
p=0.0043
p=0.0362
p=0.3048
p=0.0643
p=0.5838
p=0.4316
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2